Applied Machine Learning Lab

B.Tech Semester 4

Experiment 03

House Price Prediction using Linear Regression

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AIM

To build a machine learning model using Linear Regression to predict house prices based on various features and evaluate its performance.

THEORY

Linear Regression is a supervised learning algorithm used to model relationships between variables.

- Regression: Predicting continuous values
- Feature Scaling: Improves performance
- Model Evaluation: MAE, MSE, RMSE, R^2
- Residual Analysis: Validates model
- Feature Importance: Shows influence of features

Commands/Code Used

```
# Import libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score


# Load dataset

df = pd.read_csv("House Price India.csv")


# Preprocessing

df = df.drop_duplicates()


# Feature selection

X = df.drop("Price", axis=1)

y = df["Price"]


# Train-test split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Scaling
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)


# Model

model = LinearRegression()

model.fit(X_train, y_train)


# Predictions

y_pred = model.predict(X_test)


# Evaluation

mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, y_pred)


print("MAE:", mae)

print("MSE:", mse)

print("RMSE:", rmse)

print("R2 Score:", r2)


# Plot: Actual vs Predicted

plt.figure(figsize=(8,6))

plt.scatter(y_test, y_pred, alpha=0.5)

plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--')
```

```
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.show()

# Residuals
residuals = y_test - y_pred

plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.scatter(y_pred, residuals)
plt.axhline(0, color='red')
plt.xlabel("Predicted Price")
plt.ylabel("Residuals")
plt.title("Residuals vs Predicted")

plt.subplot(1,2,2)
sns.histplot(residuals, kde=True)
plt.title("Distribution of Residuals")

plt.show()

# Feature Importance (coefficients)
coefficients = pd.DataFrame(model.coef_, X.columns, columns=['Coefficient'])
coefficients.sort_values(by='Coefficient').plot(kind='barh', figsize=(8,6))
```

```python
plt.title("Feature Importance (Coefficients)")

plt.show()
```

RESULTS

Actual vs Predicted prices plotted.
Residual analysis performed.
Feature importance analyzed.

CONCLUSION

Linear Regression model successfully predicted house prices. Preprocessing and evaluation improved performance.